

Layered Double Hydroxide Quantum Dots for Use in a Bifunctional Separator of Lithium–Sulfur Batteries

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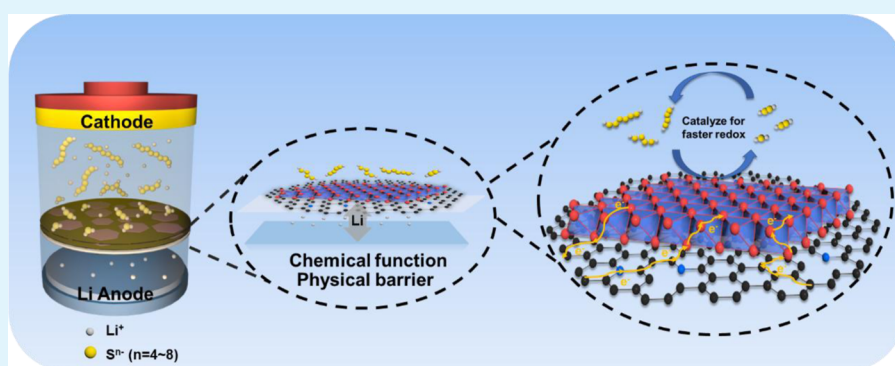
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ABSTRACT: Functional separators, which are chemically modified and coated with nanostructured materials, are considered an effective and economical approach to suppressing the shuttle effect of lithium polysulfide (LiPS) and promoting the conversion kinetics of sulfur cathodes. Herein, we report cobalt–aluminum-layered double hydroxide quantum dots (LDH-QDs) deposited with nitrogen-doped graphene (NG) as a bifunctional separator for lithium–sulfur batteries (LSBs). The mesoporous LDH-QDs/NG hybrids possess abundant active sites of Co^{2+} and hydroxide groups, which result in capturing LiPSs through strong chemical interactions and accelerating the redox kinetics of the conversion reaction, as confirmed through X-ray photoelectron spectroscopy, adsorption tests, Li_2S nucleation tests, and electrokinetic analyses of the LiPS conversion. The resulting LDH-QDs/NG hybrid-coated polypropylene (LDH-QDs/NG/PP) separator, with an average thickness of $\sim 17 \mu\text{m}$, has a high ionic conductivity of 2.67 mS cm^{-1} . Consequently, the LSB cells with the LDH-QDs/NG/PP separator can deliver a high discharge capacity of $1227.48 \text{ mAh g}^{-1}$ at 0.1C along with a low capacity decay rate of 0.041% per cycle over 1200 cycles at 1.0C .

KEYWORDS: LDH, 2D quantum dots, functional separator, redox promoter, lithium–sulfur battery

INTRODUCTION

Energy-storage devices with a high energy density and low cost beyond that of lithium-ion batteries have benefited from strong demand for sustainable renewable energy as well as emerging markets for future electronics and electric vehicles.^{1,2} Among the state-of-the-art energy-storage devices, lithium–sulfur batteries (LSBs) are the most promising candidates owing to their high theoretical energy density (2600 Wh kg^{-1}) and low cost of the sulfur cathode materials.^{3,4} However, their practical application is severely limited by the complex polysulfide conversion and the shuttle effect, which results in a sharp capacity fading, low Coulombic efficiency, and poor cyclability.^{5,6} To address this issue, tremendous efforts have been devoted to the design of cathode materials,^{7,8} such as carbon-based materials,^{9–11} metal oxides,^{12–14} carbides,^{15,16} sulfides,¹⁷ and nitrides,¹⁸ by which the shuttle effect is greatly suppressed and the reversible capacity and cycle life are effectively improved. Nonetheless, the fabrication of sulfur

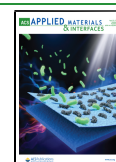
cathodes is complicated and time-consuming, and the loading of active materials is reduced, thus sacrificing the overall energy density of LSBs and posing additional challenges to their commercialization.^{19–21}

Functional separators, which are modified and coated with mesoporous carbon,^{22,23} heteroatom-doped carbon materials,²⁴ polymers,^{25,26} metal oxides,^{27,28} sulfides,²⁹ and hydroxides,^{30,31} have recently been recognized as an effective and economical approach to suppressing the shuttle effect of lithium polysulfide (LiPS) and improving the electrochemical performance of LSBs. Among these functional coating

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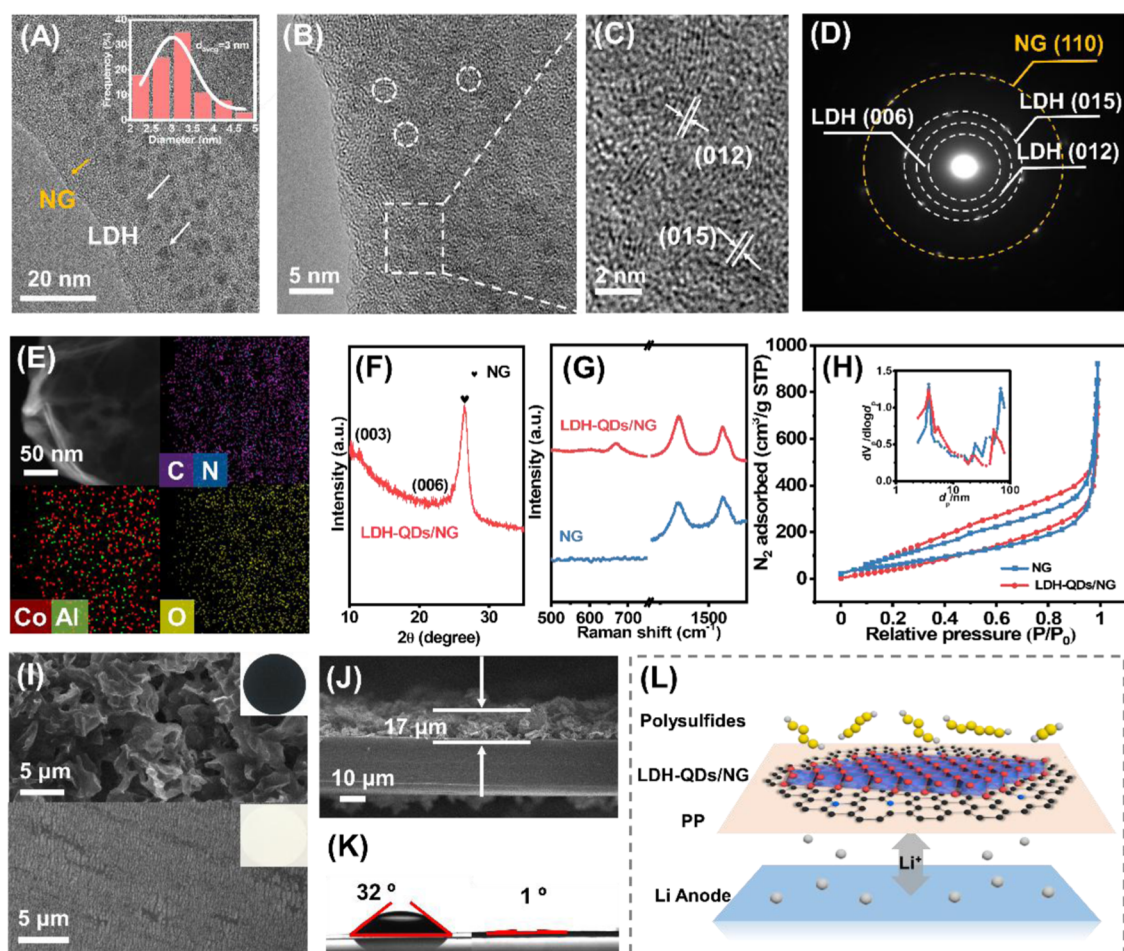


Figure 1. (A) TEM image, (B) HRTEM image, (C) enlarged HRTEM image, (D) SAED pattern, (E) EDX mapping images, and (F) XRD pattern of LDH-QDs/NG. (G) Raman spectra and (H) N_2 adsorption–desorption curves of LDH-QDs/NG and NG. (I) SEM images of LDH-QDs/NG/PP (top) and PP (bottom) separators. (J) Cross-sectional SEM image of the LDH-QDs/NG/PP separator. (K) Contact angle tests of the PP separator (left) and the LDH-QDs/NG/PP separator (right). (L) Structural diagram of the LDH-QDs/NG/PP separator with an electrolyte in the interlayer space.

materials, two-dimensional (2D) layered double hydroxide (LDH) has been considered as an intriguing candidate owing to its unique structure, large surface area, and tunable surface chemical properties.³² In particular, LDH features an ionic lamellar compound of positively charged metal hydroxide layers, which plays dual roles of effectively capturing LiPSs through polar–polar/electrostatic interactions and accelerating the redox kinetics of a sulfur cathode.^{33–35} Despite these benefits, the previously developed LDH-based coating layers are limited to large lateral sizes (hundreds of nanometers) and thicknesses (tens of nanometers), which cannot maximize the active sites for the prevention of LiPS shuttling and the promotion of conversion kinetics. The transformation of 2D layered materials into quantum dots (QDs) has been developed to achieve a high atom utilization efficiency and modified electronic properties owing to the abundant edge sites and the quantum confinement effect.³⁶ It has been demonstrated that the QD-based materials feature quantum Hall effects and enriched uncoordinated atomic structures at edge sites. Benefiting from these characteristics, the metal oxide QDs,^{37,38} carbon QDs,³⁹ and phosphorus QDs⁴⁰ have been widely studied and applied in LSBs, which can promote the chemical immobilization and conversion kinetics of LiPSs. However, the LDH-based QDs have not been employed in

LSBs. Moreover, functional separators based on LDH-QDs have yet to be explored.

Herein, we report the bifunctional coating layers of CoAl-LDH-QDs deposited onto nitrogen-doped graphene (denoted as LDH-QDs/NG) for the chemical modification of the polypropylene (PP) separator. The active sites of LDH-QDs with a small size of ~ 3 nm are exposed on the NG surface, effectively capturing LiPSs through strong polar–polar and hydrogen bonding ($O-H\cdots S$) interactions, which significantly accelerates the conversion reaction kinetics.⁴¹ Consequently, LSBs with the LDH-QDs/NG-modified separator deliver a high specific capacity of $1227.5 \text{ mAh g}^{-1}$ at $0.1C$, demonstrating long-term cycle stability with a low capacity decay of 0.04% per cycle and a high Coulombic efficiency of 99.2% over 1200 cycles at $1C$.

EXPERIMENTAL SECTION

Synthesis of CoAl-LDH Nanosheets. *CoAl-LDH* (CO_3^{2-}). To synthesize *CoAl-LDH* (CO_3^{2-}) with a well-defined hexagonal morphology, $3.5 \text{ mmol AlCl}_3 \cdot 6H_2O$, $7 \text{ mmol CoCl}_2 \cdot 6H_2O$, and 49 mmol urea were dissolved in 700 mL of deionized (DI) water. The above solution was transferred into a 1 L three-neck flask and treated at 95°C under stirring and refluxed for 48 h . The as-obtained LDHs were collected through centrifugation, washed five times with DI water and ethanol, and finally dried using lyophilization.

CoAl-LDH (Cl^-). The CoAl-LDH (Cl^-) was prepared using a salt–acid exchange method. One gram of the CoAl-LDH (CO_3^{2-}) sample was added to an aqueous salt–acid solution (1 L) with NaCl (1 mol) and HCl (3.3 mmol) in a three-neck flask. The mixture was stirred continuously at room temperature ($\sim 25^\circ\text{C}$) under a N_2 atmosphere for 24 h. The as-obtained CoAl-LDH (Cl^-) sheets were centrifuged and washed with degassed water and absolute ethanol 10 times and then dried using lyophilization.

CoAl-LDH (NO_3^-). The CoAl-LDH (NO_3^-) sheets were synthesized in the same manner as CoAl-LDH (Cl^-). The CoAl-LDH (Cl^-) sample (0.5 g) was dispersed into 0.5 L of DI water flowing with N_2 , and 0.5 g of NaNO_3 was placed in the same three-neck flask at room temperature for 24 h. After treatment, the product was collected through centrifugation, washed with ethanol and DI water five times, and finally dried using lyophilization.

LDH-QDs. Single-layer CoAl-LDH nanosheets in a colloidal suspension were obtained by dispersing 0.1 g of CoAl-LDH (NO_3^-) sheets into 100 mL of formamide and stirred for 48 h under a continuous N_2 flow. After that, the LDH-QD colloid suspension was obtained through sonication of the above suspension for 3 h.

Synthesis of Nitrogen-Doped Graphene (NG). Graphene oxide (GO) was synthesized using a modified Hummer's method.⁴² The freeze-dried GO was collocated into a 5 mg mL^{-1} suspension and evenly dispersed by sonication at room temperature. Next, 50 g of hypophosphorous acid was added into 100 mL of GO dispersion and sonicated for 30 min. Then, 5 g of iodine was added and stirred for 2 h. Subsequently, the mixture was packed into 20 mL glass vials, sealed well, and placed into an oven at 80°C for 12 h. This product was washed with DI water until the reducing agents were completely removed and the pH value reached 7, and then finally dried using lyophilization. Fifty milligrams of the as-synthesized reduced GO (rGO) was dispersed into 20 mL of DI water, whereas 100 mg of urea was added into the suspension, sonicated for 20 min, stirred for 30 min, and subsequently freeze-dried for 3 days. Then, the as-prepared powder was placed in the center of the furnace tube and heated at 900°C for 3 h with a successive N_2 flow. After that, the final product of NG was obtained through the ozone treatment for 30 min.

Synthesis of the LDH-QDs/NG-Modified PP Separator. The LDH-QDs/NG composite was prepared through the self-assembly of LDH-QDs and NG sheets. Here, 5 mg of NG was dispersed and collocated into a 50 mL *N*-methyl-2-pyrrolidone (NMP) solvent through a 3 h sonication. The as-prepared CoAl-LDH-QD colloid suspension was added into the above solvent under magnetic stirring. In addition, poly(vinylidene fluoride) (PVDF) as a binder with a weight ratio of 10% was added to the mixed suspension under stirring for another 2 h. Subsequently, the mixture was vacuum-filtered and coated onto a PP separator (Celgard 2400). After coating, the LDH-QD/NG-modified separator was dried in a vacuum oven at room temperature for 12 h. The resultant LDH-QDs/NG/PP was punched into disks with a diameter of 18 mm and a mass loading of $\sim 0.3\text{ mg cm}^{-2}$ for the fabrication of LSB cells.

RESULTS AND DISCUSSION

The LDH-QDs were synthesized through the intercalation and delamination of CoAl-LDH nanosheets, followed by subsequent ultrasonication treatment processes (see the [Experimental Section](#) for details). First, the as-exfoliated CoAl-LDH nanosheets were synthesized by exchanging the interlayer from CO_3^{2-} of CoAl-LDH through Cl^- to NO_3^- and the delamination process in formamide ([Figures S1 and S2](#)). After a long-time ultrasonication treatment (3 h), the LDH-QD colloid suspension was obtained with a clear Tyndall effect shown in [Figure S3](#). Subsequently, the as-synthesized LDH-QDs were self-assembled onto the ozone-treated NG ([Figure S4](#)) through electrostatic adsorption.

Finally, the LDH-QDs/NG-modified PP separator was obtained using a simple vacuum filtration process ([Figure S5](#)).

The morphology of the LDH-QDs/NG hybrids was examined using transmission electron microscopy (TEM). As shown in [Figures 1A and S6](#), numerous LDH-QDs are uniformly distributed on the NG surface without any aggregation. The magnified TEM image ([Figure 1B](#)) reveals that the average size of the LDH-QDs is approximately 3 nm, which is crucial for the efficient utilization of active sites arising from the acceleration of polysulfide redox kinetics. Thus, the uniform deposition of LDH-QDs onto the conductive NG surface is expected to provide a fast electron transfer and abundant active sites during the redox reaction processes. From the high-resolution transmission electron microscopy (HRTEM) image of LDH-QDs/NG hybrids shown in [Figure 1C](#), lattice fringes of 0.26 and 0.23 nm, representing the (012) and (015) planes, can be observed, indicating the characteristic crystalline structure of the LDH-QDs. The typical selected-area electron diffraction (SAED) pattern in [Figure 1D](#) represents the mixed dot and ring patterns, corresponding to the (006), (012), and (015) facets of LDH-QDs and the (110) facet of NG, respectively. Energy-dispersive X-ray (EDX) mapping proves the composition of LDH-QDs/NG hybrids, showing a uniform distribution of the C, N, Co, Al, and O elements on the NG surface throughout the entire detected area ([Figures 1E and S7](#)).

The crystallographic and microstructure of the LDH-QDs/NG hybrids were further analyzed using X-ray diffraction (XRD) and Raman spectroscopy. As shown in [Figure 1F](#), a typical diffraction peak of NG is captured at 26.2° . In particular, very weak peaks at 10.2 and 23.2° , corresponding to the (003) and (006) planes, are observed. The relatively weak intensity is mainly due to the ultrafine structure of the LDH-QDs ($\sim 3\text{ nm}$), as confirmed by the HRTEM results. Raman spectra confirm that a new band of LDH-QDs/NG hybrids appeared at 676 cm^{-1} ([Figure 1G](#)), which is associated with the M–O–M or M–OH (M = Co and Al) linkage bands, further confirming the presence of LDH-QDs in the hybrids. In addition, the Raman spectrum of the LDH-QDs/NG hybrids shows two bands at 1346 and 1569 cm^{-1} , representing the D and G bands of graphene. The intensity ratio of the LDH-QDs/NG hybrids ($I_D/I_G = 1.40$) is higher than that of NG ($I_D/I_G = 1.27$), suggesting the anchoring of LDH-QDs on the NG surface. The LDH-QDs/NG hybrids feature a large specific surface area of $242\text{ m}^2\text{ g}^{-1}$ with multilayer adsorption, as analyzed by the N_2 sorption/desorption isotherm ([Figure 1H](#) and [Table S1](#)). The mesoporous structure, which is attributed to the hysteresis of the isotherm, provides abundant active sites and transporting channels to accelerate the redox kinetics of the LiPSs.

The LDH-QDs/NG hybrids were readily coated onto the PP surface through a vacuum filtration process, resulting in a modified LDH-QDs/NG/PP separator. The top-view SEM image shows the uniform coating of LDH-QDs/NG hybrids onto the PP surface, where the pores of the PP are not blocked but are fully covered ([Figure 1I](#)). The cross-sectional SEM image further reveals the homogeneous coverage of LDH-QDs/NG hybrids with a thickness of $\sim 17\text{ }\mu\text{m}$ ([Figure 1J](#)). The mesoporous structure of the LDH-QDs/NG is preserved after coating onto the PP surface. It should be noted that the LDH-QDs/NG coating layers can improve the hydrophilicity of the PP surface, as verified by the contact angle test ([Figure 1K](#)). Benefiting from the enhanced

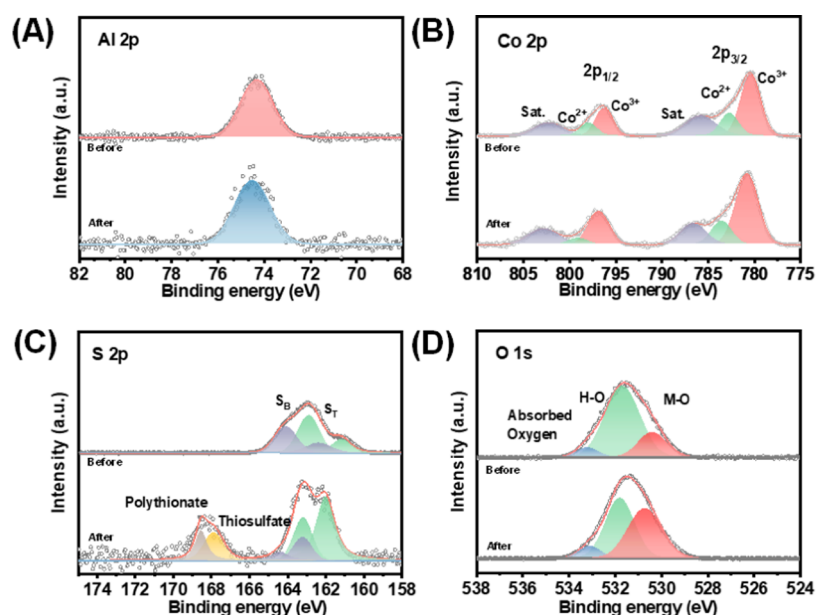


Figure 2. The XPS spectra of LDH-QDs/NG hybrids (before and after absorption with Li_2S_6): (A) Al 2p, (B) Co 2p, (C) S 2p, and (D) O 1s.

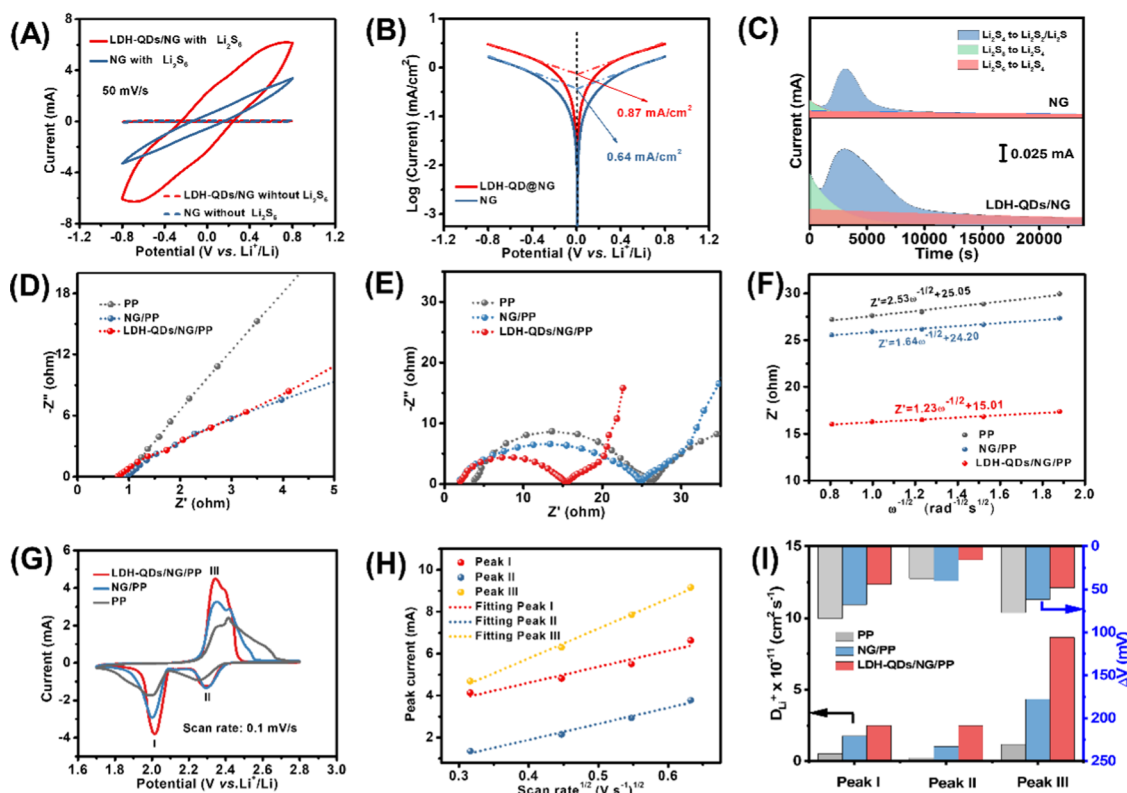


Figure 3. (A) CV curves and (B) Tafel plots of symmetric cells with NG and LDH-QDs/NG hybrids at 50 mV s^{-1} . (C) Potentiostatic discharge curves representing Li_2S_6 nucleation on the surface of NG and LDH-QDs/NG. (D) Nyquist plots of the symmetric cells with pristine PP, NG/PP, and LDH-QDs/NG/PP separators between two stainless steel electrodes. (E) Nyquist plots of the LSB cells with pristine PP, NG/PP, and LDH-QDs/NG/PP separators after 10 cycles at 1C. (F) Relationship between the real part of impedance and low frequencies. (G) CV curves of the LSB cells with PP, NG/PP, and LDH-QDs/NG/PP. (H) Relationship between the scan rate $^{1/2}$ to peak current. (I) Li-ion diffusion coefficient (black axis) and polarized voltage (blue axis) comparison for the LSB cells with PP, NG/PP, and LDH-QDs/NG/PP separators.

wettability, the penetration of Li ions across the LDH-QDs/NG/PP separator and their access to the active sites are facilitated during the redox reaction process (Figure 1L), further boosting the electrochemical performance of the LSB cells.

To reveal the adsorption and chemical interactions between the LiPSs and LDH-QDs/NG coating layers, adsorption experiments of LDH, NG, and LDH-QDs/NG hybrids toward Li_2S_6 were conducted under identical conditions. The bulk absorptivity is visually observed in the order of LDH-QDs/

NG > LDH > NG (Figure S8), indicating the largest available surface polar sites and hydroxyl groups of the LDH-QDs/NG.

The chemical adsorption of LiPSs by LDH-QDs/NG was further investigated by X-ray photoelectron spectroscopy (XPS). As shown in Figure 2A,B, compared with the pristine sample, the Al and Co peaks shift to a marginally higher binding energy after the adsorption of Li_2S_6 , indicating a strong chemical affinity between Li_2S_6 and LDH-QDs. In addition, the peaks are broadened, disclosing different chemical environments at the center of the Al and Co metal ions owing to the interaction with Li_2S_6 . For the Co 2p region, two strong peaks are observed at 780.40 and 796.25 eV for the LDH-QDs/NG hybrids, which are assigned to $2p_{3/2}$ and $2p_{1/2}$, respectively (Figure 2B). This finding suggests the coexistence of Co^{2+} and Co^{3+} ions.⁴³ In addition, two broad satellite peaks (indicated as "Sat.") are observed at 786.61 and 802.23 eV, which are attributed to the Co–OH bond.^{44,45} After soaking LDH-QDs/NG hybrids in the Li_2S_6 solution, the relative contributions of Co^{3+} and Co^{2+} ions to the overall Co intensity are changed, as verified by the increase in the relative $\text{Co}^{3+}/\text{Co}^{2+}$ ratio of $2p_{3/2}$ and $2p_{1/2}$ from 2.78 to 5.77 and 2.96 to 3.34, respectively. The partial increase of the valence state for Co species indicates their involvement in the chemical adsorption of LiPSs. As shown in Figure 2C, the peaks in the high-resolution S 2p spectrum of LDH-QDs/NG- Li_2S_6 reflect four sulfur environments. The double peaks at 163.2 and 164.4 eV are assigned to the bridging sulfur (S_B) of Li_2S_6 , whereas the double peaks at 161.9 and 163.1 eV are associated with the terminal sulfur (S_T) of Li_2S_6 . Moreover, the peaks at 167.9 and 168.6 eV correspond to thiosulfate and polythionate species, respectively, resulting from the oxidation of Li_2S_6 . Compared to the pristine Li_2S_6 , two peaks positively shift by 0.7 eV for S_B and 0.3 eV for S_T , resulting from the interaction between Li_2S_6 and LDH-QDs.^{46,47} As indicated in Figure 2D, the O 1s peaks at 530.4, 531.7, and 533.1 eV are assigned to lattice O^{2-} (M–O), the hydroxyl species (H–O), and absorbed oxygen, respectively. After the adsorption of Li_2S_6 , the relative area ratios of the two peaks changed considerably from 20.52 to 45.50% for M–O and from 73.25 to 47.54% for H–O, along with a decrease in the area ratio of H–O/M–O from 3.57 to 1.05. Considering that both O^{2-} and the surface hydroxyl species are active oxygen species and play a significant role in oxidation reactions;^{48–50} these results indicate that the surface hydroxyl species of LDH-QDs are involved in the chemical adsorption and reaction with LiPSs.

To elucidate the redox kinetics of LiPSs on the LDH-QDs/NG surface, symmetric 2032 coin cells were assembled (see the Supporting Information Experimental Section). As depicted in Figure 3A, the LDH-QDs/NG shows a higher current density than NG, clarifying the more facile redox kinetics of Li_2S_x conversion to low-grade LiPSs. Nearly identical CV curves are also observed at different cycles, implying the facile redox kinetics of the LDH-QDs/NG at high rates (Figure S9). The facile conversion kinetics of the LDH-QDs/NG surface is further confirmed by the larger exchange current density (0.87 mA cm^{-2}) than that of the NG surface (0.64 mA cm^{-2}), as derived from the Tafel plots (Figure 3B). Moreover, the nucleation and growth of Li_2S were investigated through potentiostatic discharge experiments (see the Supporting Information Experimental Section). Figure 3C shows the precipitation of Li_2S on the surface of NG and LDH-QDs/NG. It should be noted that during the initial incubation period, the decreased current is

attributed to the reduction of higher-order LiPSs to Li_2S_4 . The LDH-QDs/NG shows a shorter incubation time than that of NG, illustrating the fast conversion kinetics for the reduction into Li_2S . Meanwhile, the specific capacity of LDH-QDs/NG for the Li_2S nucleation is $194.98 \text{ mAh g}^{-1}$, which is much higher than 79.70 mAh g^{-1} of NG.

The effect of the LDH-QDs/NG coating layers on the Li-ion conductivity of the modified separator was investigated using electrochemical impedance spectroscopy (EIS). Nyquist plots of the symmetrical LSB cells with the same electrolyte between two stainless steel electrodes were collected, as shown in Figure 3D. The intercept with the x -axis of each plot indicates the ionic resistance of the corresponding separators. The Li-ion conductivity of the LDH-QDs/NG/PP separator is 2.67 mS cm^{-1} , which is higher than those of the PP separator (1.31 mS cm^{-1}) and NG/PP separator (2.20 mS cm^{-1}). This finding indicates that the LDH-QDs/NG coating layer slightly increases the Li-ion conductivity owing to the porous ion channels and hydrophilic functional groups despite the full coverage of the pore surfaces of the pristine PP separator.

The facile and reversible redox kinetics of the LSBs with the LDH-QDs/NG/PP separator, MWCNT/S cathode, and Li anode were investigated using Nyquist plots within the frequency range of 10^6 – 10^{-1} Hz. As shown in Figure 3E, the charge-transfer resistance ($R_{\text{ct}} = 15.40 \Omega$) of the LSB cell with the LDH-QDs/NG/PP separator is lower than that of the NG/PP separator ($R_{\text{ct}} = 24.77 \Omega$) and PP separator ($R_{\text{ct}} = 26.14 \Omega$). This means that the charge transfer at the interface between the LDH-QDs/NG and LiPSs is facilitated for the improved conversion kinetics of the LiPSs. In addition to the charge-transfer kinetics, ion diffusion into the active sites of the electrode is a key parameter for determining the facile redox kinetics. Figure 3F illustrates the linear relationship between the real impedance (Z') and frequency ($\omega^{-1/2}$) for the cells with LDH-QDs/NG/PP, NG/PP, and PP separators within the low-frequency region. The slope of this plot indicates the degree of solid-state diffusion of Li^+ ions in the internal electrode.⁵¹ The fitted slope of the LDH-QDs/NG/PP separator-based cell is 1.23, which is smaller than those of the others, demonstrating the fast ion diffusion kinetics. To demonstrate the redox promoting effect of LDH-QDs/NG, the CV curves of LSBs with different separators were measured at 0.1 mV s^{-1} . As shown in Figure 3G, CV curves display two cathodic peaks at 1.9–2.1 V(I) and 2.2–2.4 V(II) during the discharging process, corresponding to the reduction of S_8 into soluble long-chain Li_2S_x ($4 < x < 8$) and long-chain Li_2S_x ($4 < x < 8$) into insoluble $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$, respectively. Upon charging, an anodic peak is observed at approximately 2.4–2.6 V(III), representing the oxidation of $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$ into S_8 . Compared to the PP and NG/PP separators, the LDH-QDs/NG/PP separator displays a positive shift of the cathodic peaks and a negative shift of the anodic peak with higher current intensities. During the subsequent cycles, the cathodic and anodic peaks with the LDH-QDs/NG/PP separator are stabilized (Figure S10). These results imply that the LDH-QDs/NG coating layer could significantly boost the reversibility and redox kinetics of the LiPSs.

To further explore the superiority of the LDH-QDs/NG-modified layer during the enhanced redox reactions, the lithium-ion coefficients (D_{Li^+}) were calculated based on the Randles–Sevcik equation.

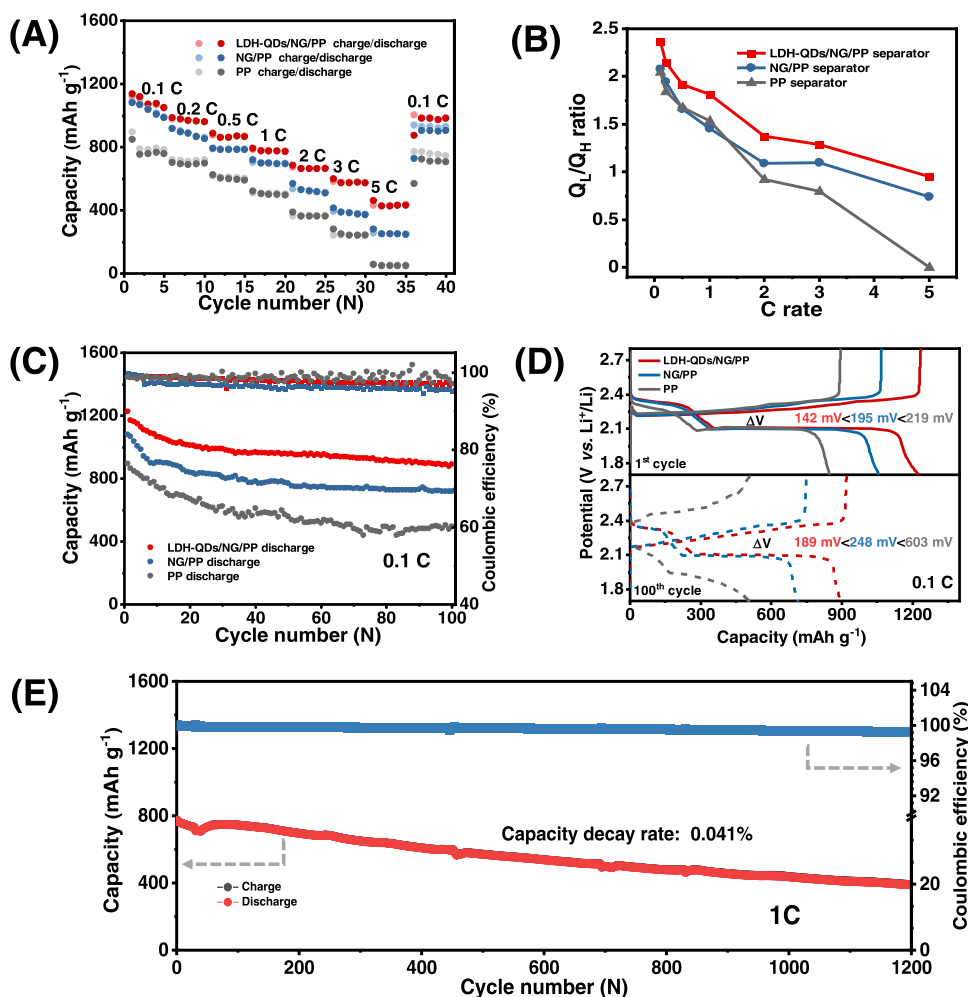


Figure 4. (A) Specific capacities and (B) Q_L/Q_H ratios of LSBs with PP, NG/PP, and LDH-QDs/NG/PP separators at various C-rates. (C) Cycling performance and (D) charge–discharge curves of LSBs with PP, NG/PP, and LDH-QDs/NG/PP separators at 0.1C; (E) long cycle performance of LSB with the LDH-QDs/NG/PP separator over 1200 cycles at 1C.

$$I_p = 2.69 \times 10^5 n^{1.5} A D_{Li^+}^{0.5} C_{Li^+}^{0.5} \nu^{0.5}$$

where I_p is the peak current (A), n is the electron transfer number ($n = 2$), A is the electrode area (1.77 cm^2), C_{Li^+} is the Li^+ ion concentration in the electrolyte (mol cm^{-3}), D_{Li^+} is the diffusion coefficient of Li ions ($\text{cm}^2 \text{ s}^{-1}$), and ν is the scan rate (V s^{-1}).⁵² This equation indicates a linear relationship between I_p and $\nu^{0.5}$ when n , A , and C are constant. Figure 3H shows the peak current variation corresponding to the square root of the scan rates and the corresponding CV profiles, as displayed in Figure S11. As shown in Figure 3I, the D_{Li^+} of the cells with LDH-QDs/NG/PP separators possess the largest peaks I ($1.18 \times 10^{-11} \text{ cm}^2 \text{ s}^{-1}$), II ($4.33 \times 10^{-11} \text{ cm}^2 \text{ s}^{-1}$), and III ($8.64 \times 10^{-11} \text{ cm}^2 \text{ s}^{-1}$), whereas the lowest peak-to-peak separation (ΔV) occurs in peaks I (37 mV), II (16 mV), and III (40 mV) among the three LSB cells (calculated according to the data from Figure S11). These findings confirm that the LDH-QDs/NG/PP separator has a faster Li-ion diffusion and charge-transfer rate as well as higher Li-ion conductivity, finally promoting the redox conversion kinetics.

Based on the above results, to illustrate the strong capture ability of the LiPSs and the remarkable conversion kinetics toward the LiPSs of the LDH-QDs/NG/PP separators on the electrochemical performance, 2032-type coin cells were assembled with a modified separator using MWCNT/S as

the cathode and lithium metal as the anode. The content of sulfur is $\sim 77\%$, as characterized through thermogravimetric analysis (Figure S12). As shown in Figure 4A, the cell with the LDH-QDs/NG/PP separator could deliver discharge capacities of 1134.26, 983.86, 884.25, 790.50, 683.10, 595.89, and 458.46 mAh g^{-1} at current densities of 0.1, 0.2, 0.5, 1.0, 2.0, 3.0, and 5.0C, respectively. When the current density is reset to 0.1C, the reversible capacity can be sustained at 980.97 mAh g^{-1} . In comparison, the cells with PP and NG/PP separators only deliver initial discharge capacities of 847.51 and 1080.19 mAh g^{-1} , respectively, with fast capacity fading (Figure S13). The ratios of the low plateau region discharge capacity (Q_L) and the high plateau region discharge capacity (Q_H) were further calculated based on the discharge–charge curves (Figure S13).⁵³ As shown in Figure 4B, at current densities from 0.1 to 5C, the values of the Q_L/Q_H ratio for the LDH-QDs/NG/PP separator are higher than those of the reference samples at different current densities, suggesting the facile and reversible conversion redox kinetics and mitigated polysulfide diffusion of the LDH-QDs/NG/PP separator.

Another metric reflecting the reversible electrode kinetics is the cycle performance of the LSB cells. We compared the cycling performances of LSB cells with the LDH-QDs/NG/PP, NG/PP, and PP separators. As shown in Figure 4C, the LSB cell with the LDH-QDs/NG/PP separator reaches an

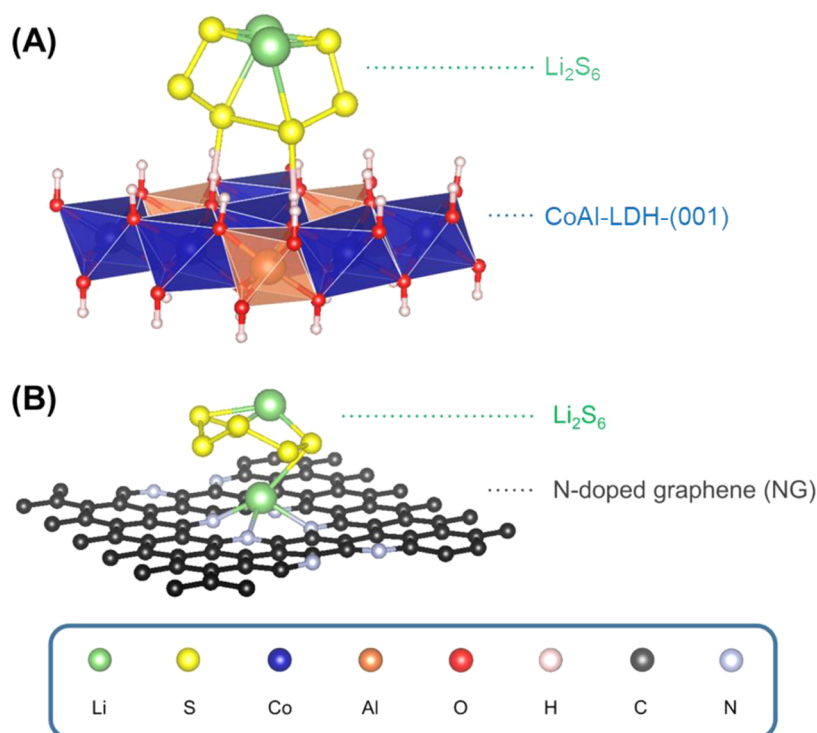


Figure 5. Calculated Li_2S_6 -adsorption structure of (A) LDH and (B) NG.

initial discharge capacity of $1227.48 \text{ mAh g}^{-1}$ at 0.1C. A stable capacity of $894.52 \text{ mAh g}^{-1}$ is retained after 100 cycles with a high Coulombic efficiency of 97.17% and a low cycle decay ratio of 0.27%. By contrast, an LSB cells with a PP separator and a NG/PP separator display low discharge capacities of 505.14 and $713.78 \text{ mAh g}^{-1}$, respectively, with lower Coulombic efficiencies of 96.95 and 95.02% and higher cycle decay ratios of 0.41 and 0.33%, respectively. Interestingly, among the three separators, the smallest overpotential for the LDH-QDs/NG/PP separator still remained at both the 1st and 100th cycles (Figure 4D). This suggests that the LDH-QDs/NG/PP separator can improve the redox kinetics and decrease the internal resistance. In addition, the morphology and electronic structure of LDH-QDs/NG are well maintained after the cycling test (Figures S14–S16). The long-term cycling performance of the LSB cell with the LDH-QDs/NG-modified separator was further confirmed, as displayed in Figure 4E. The LSB cell with the LDH-QDs/NG/PP separator delivers an initial discharge capacity of $804.78 \text{ mAh g}^{-1}$ at 1C. After 1200 cycles, it remains at a low capacity fading rate of 0.041% per cycle with a Coulombic efficiency of 99.27% at 1C. The electrochemical performance of the LSBs was compared with those of previous studies regarding modified separators (Table S2). This remarkable cycling performance is attributed to the strong chemical affinity between LDH-QDs/NG and LiPSs and the promoted conversion redox kinetics for the effective inhibition of LiPS shuttling. In addition, to demonstrate the suppressed shuttle effect of LiPSs by LDH-QDs and NG, we performed density functional theory (DFT) calculation on theoretical absorption energies (E_{ads}) of Li_2S_6 at the LDH (001) surface and NG. Figure 5 presents the theoretically predicted Li_2S_6 -adsorbed LDH and NG structures. The Li_2S_6 -absorption energies of LDH and NG are -2.31 and -1.91 eV , respectively (Table

S3), suggesting that LDH-QDs can deliver more enhanced Li_2S_6 -absorption ability compared to NG. These DFT calculation results are well matched with the experimental results showing that LDH-QDs/NG exhibits more effectively inhibited shuttling of LiPS than NG. Remarkably, the LSB cells using the LDH-QDs/NG/PP separator operated at higher sulfur loading (3.0 and 3.6 mg cm^{-2}) and a low electrolyte concentration ($9.8 \mu\text{L mg}^{-1}$) still delivered a high reversible specific capacities of 892 and 859 mAh g^{-1} after 25 cycles (Figure S17). This research provides a chemical strategy for the rational design of bifunctional separators using LDH-QDs to inhibit polysulfide shuttling effects and promote the redox kinetics for high-performance LSB cells.

CONCLUSIONS

In summary, we designed LDH-QDs assembled with NG coated on a commercial separator to promote the electrochemical performance of the LSBs. The straightforward modification of the PP separator allows the integration of a conductive carbon layer, which offers physical appeal to tie dissolved polysulfide and fast electron transfer for a redox reaction. In addition, the modified separator can impose restrictions on the active material on the cathodic side to facilitate its reutilization. As a consequence, the LSB with a LDH-QDs/NG-coated separator achieves an increased performance, i.e., high capacity, long cycle life, and low capacity fading at different C rates. Our results show the significance of a rational design structure and the potential of using LDH-QDs/NG as modified separators, which can improve the performance of advanced high-performance LSBs.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.1c00974>.

The schematic illustration of materials synthesis and detailed experimental procedures and experimental data including SEM, TEM, BET, XRD, XPS, and electrochemical data (PDF)

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Notes

The authors declare no competing financial interest.

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